

Generation Interconnection System Impact Study Report

For Queue Project Q13-0242

Cs Murphy Point Llc 138 kV

August 26, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0242. The Interconnection Customer proposes a 38.1 MW solar photovoltaic facility interconnecting at the Cs Murphy Point Llc 138 kV substation in the Virginia area. The request was received on June 01, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 179 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Westport-Wheelabrator Baltimore Refuse 138 kV is overloaded to 114.7% of its 197 MVA emergency rating for the loss of the Taber-Dixleta 345 kV. The project contributes 6.4% DFAX (2.5 MW) to this overload.

The Sunbright-Fairlad 138 kV line is overloaded to 109.9% of its 406 MVA emergency rating for the loss of the Taber-Glnannan 345 kV line. The project contributes 8.3% DFAX (3.2 MW) to this overload.

For the loss of Taber-Glenannan 345 kV, the Sunbright-Colesville 138 kV loads to 106.8% of its 470 MVA rating. Project DFAX: 8.3%.

Dixleta-Drum Substation 138 kV: 106.1% loading (428 MVA) following the outage of the Taber-Dixleta 345 kV; DFAX = 6.2%.

The Fairland-Siuc 138 kV is overloaded to 105.1% of its 340 MVA emergency rating for the loss of the Taber-Glenannan 345 kV. The project contributes 5.5% DFAX (2.1 MW) to this overload.

Jerome-High Ridge 138 kV: 104.2% loading (358 MVA) following the outage of the Taber-Dixleta 345 kV; DFAX = 9.2%.

The Wheelabrator Baltimore Refuse-Wilkens 138 kV is overloaded to 103.5% of its 314 MVA emergency rating for the loss of the Dixleta-Taber 345 kV L. The project contributes 6.3% DFAX (2.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wheelabrator Baltimore Refuse-Westport 138 kV	Rebuild line with higher-capacity conductor	\$582,000	1% (\$4,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sunbright-Fairland 138 kV line	Reconductor line	\$180,000	1% (\$1,000)
Sunbright-Colesville 138 kV	Reconductor line	\$699,000	1% (\$4,000)
Dixleta-Drum Substation 138 kV	Reconductor line	\$780,000	1% (\$5,000)
Fairland-Siuc 138 kV	Reconductor line	\$550,000	1% (\$5,000)
Jerome-High Ridge Substation 138 kV	Reconductor line	\$828,000	1% (\$5,000)
Wilkens-Wheelabrator Baltimore Refuse Substation 138 kV	Reconductor line	\$442,000	1% (\$3,000)
	Total allocated to project		\$27,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.